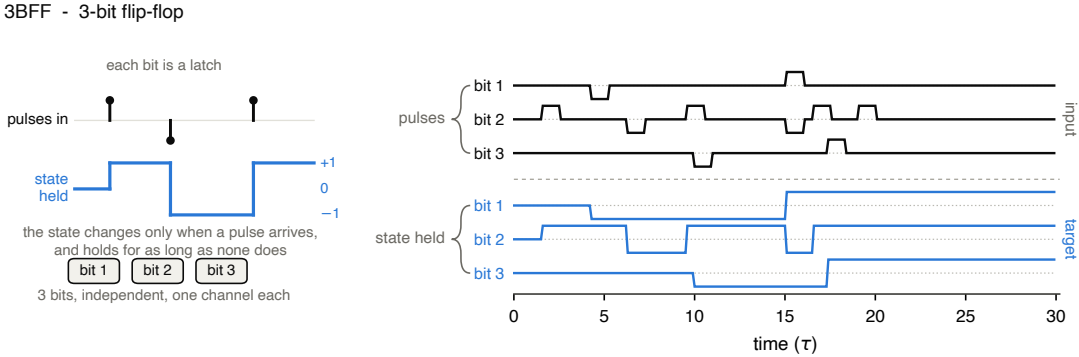
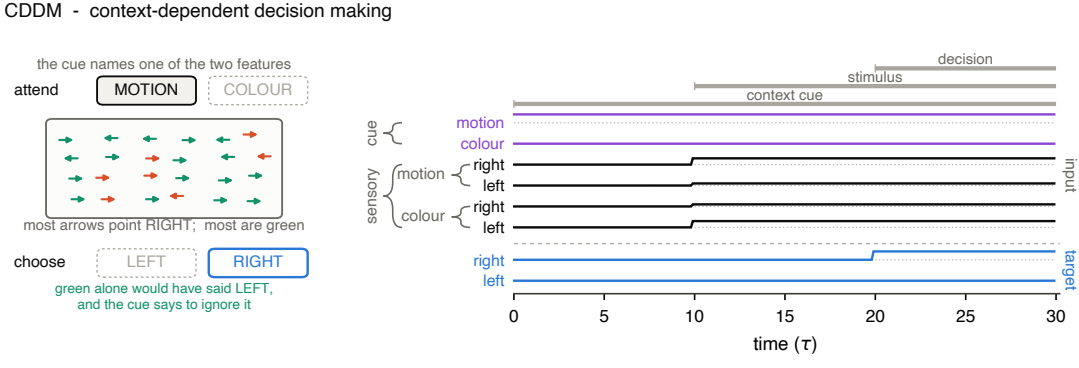
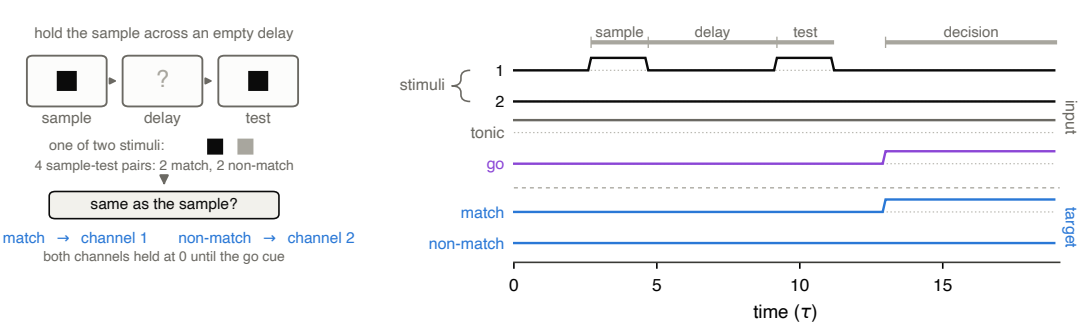
a


3BFF - 3-bit flip-flop
each bit is a latch
pulses in
state held
+1
0
−1
the state changes only when a pulse arrives,
and holds for as long as none does
bit 1
bit 2
bit 3
3 bits, independent, one channel each
pulses
bit 1
bit 2
bit 3
input
state held
bit 1
bit 2
bit 3
target
0
5
10
15
20
25
30
time (τ)

b


CDDM - context-dependent decision making
the cue names one of the two features
attend
MOTION
COLOUR
most arrows point RIGHT; most are green
choose
LEFT
RIGHT
green alone would have said LEFT,
and the cue says to ignore it
decision
stimulus
context cue
cue
motion
colour
sensory
motion
right
left
colour
right
left
input
right
left
target
0
5
10
15
20
25
30
time (τ)

c


DMTS - delayed match-to-sample
hold the sample across an empty delay
?
sample
delay
test
one of two stimuli:
4 sample-test pairs: 2 match, 2 non-match
same as the sample?
match → channel 1
non-match → channel 2
both channels held at 0 until the go cue
sample
delay
test
decision
stimuli
1
2
tonic
go
input
match
non-match
target
0
5
10
15
time (τ)